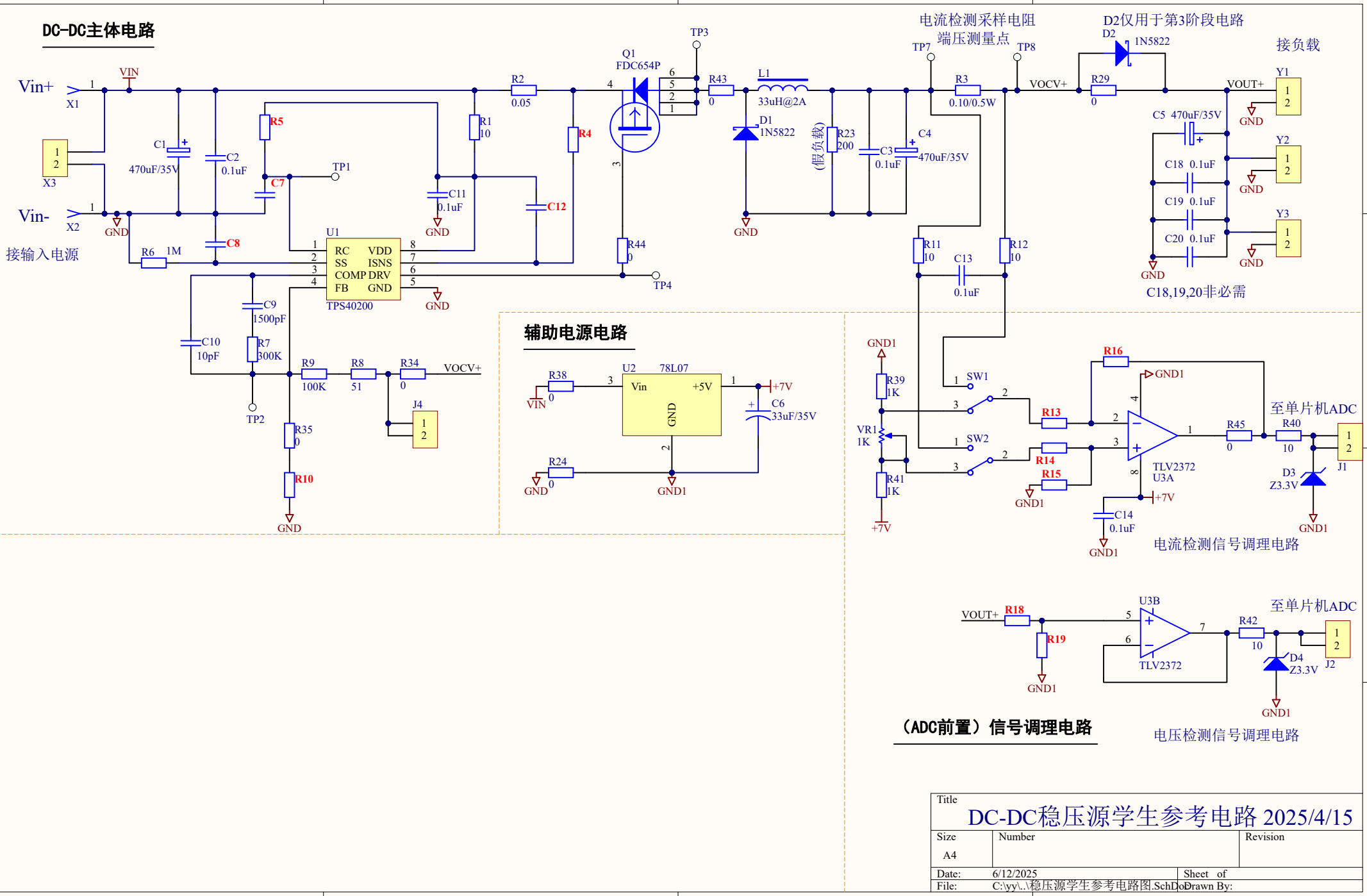

DC-DC主体电路
Vin+
Vin-
接输入电源
X1
X2
X3
VIN
GND
C1
470uF/35V
C2
0.1uF
C7
R5
TP1
R6
1M
C8
U1
RC
SS
COMP
FB
VDD
ISNS
DRV
GND
TPS40200
C9
1500pF
C10
10pF
R7
300K
R9
100K
R8
51
R34
0
VOCV+
J4
TP2
R35
0
R10
R1
10
C11
0.1uF
C12
R2
0.05
R4
Q1
FDC654P
TP3
R43
0
L1
33uH@2A
D1
1N5822
R44
0
TP4
(假负载)
R23
200
C3
0.1uF
C4
470uF/35V
电流检测采样电阻
端压测量点
TP7
TP8
R3
0.10/0.5W
VOCV+
D2仅用于第3阶段电路
D2
1N5822
R29
0
VOUT+
接负载
Y1
Y2
Y3
C5 470uF/35V
C18 0.1uF
C19 0.1uF
C20 0.1uF
C18,19,20非必需
R11
10
C13
0.1uF
R12
10
辅助电源电路
U2
78L07
Vin
+5V
GND
R38
0
R24
0
C6
33uF/35V
+7V
GND1
GND1
R39
1K
VR1
1K
R41
1K
+7V
SW1
SW2
R16
R13
R14
R15
TLV2372
U3A
R45
0
R40
10
至单片机ADC
J1
D3
Z3.3V
C14
0.1uF
电流检测信号调理电路
VOUT+
R18
R19
U3B
TLV2372
R42
10
至单片机ADC
J2
D4
Z3.3V
电压检测信号调理电路
（ADC前置）信号调理电路
Title
DC-DC稳压源学生参考电路 2025/4/15
Size
A4
Number
Revision
Date: 6/12/2025
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File: C:\yy\..\稳压源学生参考电路图.SchDoc
Drawn By: